U.S. ARMY COMBAT READINESS CENTER

HUMAN FACTORS ANALYSIS AND CLASSIFICATION SYSTEM (HFACS) GUIDEBOOK

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Introduction

Human error remains the leading cause of Army mishaps. On-duty mishaps and near-misses are rarely attributable to a single cause or one person's unsafe act. Instead, mishaps are the result of a series of long-standing flaws defined as system inadequacies. These system inadequacies reside throughout multiple levels of an organization and are often a result of gaps in training, resource support, leader actions, individual actions, and/or policy and procedures.

The Investigation Division of the USACRC developed this guide as a desk-side supplement to DA PAM 385-40. The Human Factors Analysis and Classification System (HFACS) will assist safety personnel and members of a Safety Investigation Board (SIB) in explaining the causal factors related to human factors and aid in capturing human error codes for trend analysis.

The Army HFACS model incorporates the three tiers of the traditional HFACS model (preconditions, supervisory influences, and organizational influence) into five **system inadequacy (SI)** categories better suited for US Army application. The five SI categories are:

1. **Individual**
2. **Leader**
3. **Support**
4. **Training**
5. **Standards**

Introduction

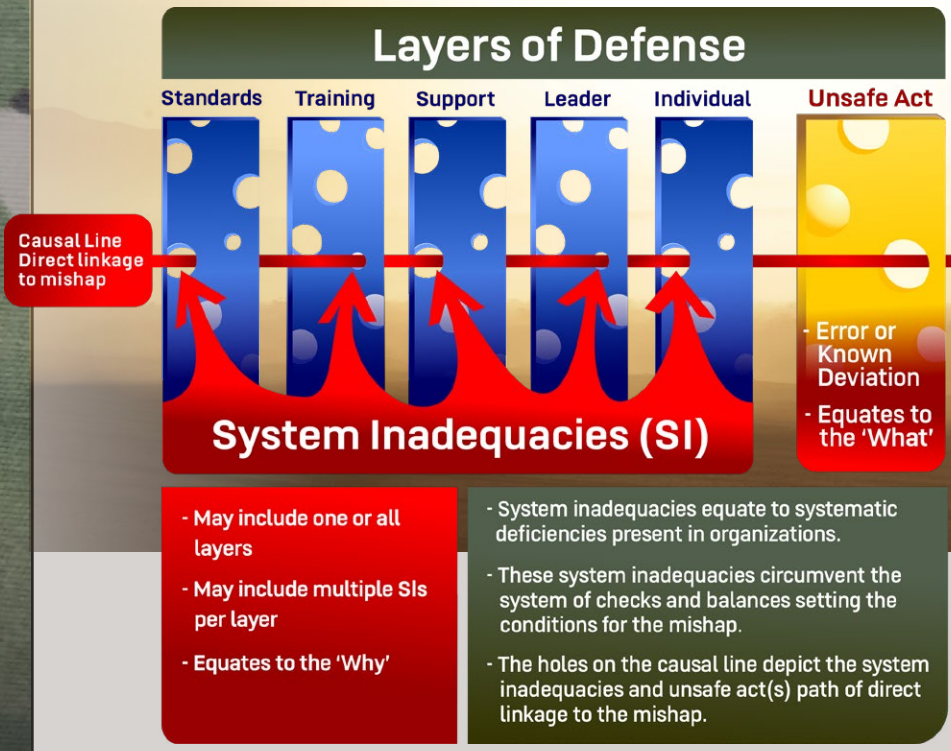

Figure 1: Swiss Cheese Model (Reason's Causation Model)

Each of these five categories contain codes for conditions or 'system inadequacies' that support the Investigation's efforts to determining 'why' the mishap occurred. These system inadequacies or SIs are long-term unassessed, ineffectively assessed, or undetected hazardous conditions that result in individual unsafe acts.

TIP: SIs identify 'why the mishap person committed the unsafe act' or 'why the mishap occurred.'

Unsafe Acts are the ***actions committed by an operator or mishap person that results in the execution of a human error or unsafe situation.*** The actions are closely related or tied directly to the operator or mishap person.

There are two distinct categories of unsafe acts: (1) errors and (2) known deviations.

*The error category includes two subcategories: (1) performance-based errors and (2) judgement or decision-making errors.

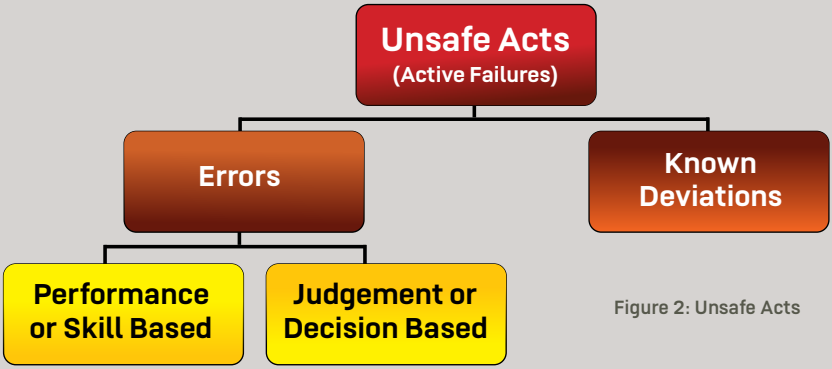

Figure 2: Unsafe Acts

How to use the DoD HFACS tool

Step 1: Determine all factors that caused the mishap.
The SIB must identify all causal factors before assigning the HFACS codes. SIBs must determine not only unsafe acts or "what happened," but most importantly, the system inadequacies as to "why the mishap occurred." Once the SIB conducts analysis and identifies causal factors, place each causal factor into its respective category of human, materiel, and environmental factors (environmental factors and material factors are not coded using HFACS, see DA PAM 385-40 for the appropriate tables). The causal factors form the basis for the findings that are contributing to the mishap, severity of injury, or severity of damage. The SIB needs to record the codes for entry into ASMIS 2.0. This handbook IS NOT the best technique for use during deliberations. The SIB should use the deliberations tools located at **https://safety.army.mil/REPORTING-INVESTIGATION/Tools**.

Step 2: Identify and assign HFACS codes to unsafe acts.
Based on analysis of causal factors, first identify the unsafe act(s) (what happened). Unsafe acts are typically individual actions taken that directly resulted in the mishap. There are no limits to the number of unsafe acts, however, **there is only one unsafe act per finding** (see DA PAM 385-40 for information related to the 7 elements of a finding). Unsafe acts fall into 3 categories, (1) judgment/decision-based errors, (2) performance-based errors, and (3) known deviations. Answer the questions preceding each unsafe act bin, then capture the most applicable code relating to the corresponding causal factor. Record the applicable code on the deliberation worksheet located at **https://safety.army.mil/REPORTING-INVESTIGATION/Tools**. Once the SIB codes the

unsafe act, move to the identification and coding of the system inadequacies related to the unsafe act.

NOTE: **If the SIB identifies multiple unsafe acts codes, apply system inadequacies starting with the first unsafe act. Once complete, move to the next unsafe act and repeat the process.**

Step 3: Identify and assign HFACS codes to System Inadequacies (SIs). After coding the unsafe act, code causal factors related to why the person committed the unsafe act using the SI HFACS codes. SIs relate to the "why" there was an unsafe act. After identifying a causal factor related to why there was an unsafe act, move to the human error flow chart (figure 3) and apply the questions to the "why" causal factor. Repeat this process for each causal factor related to the "why" there was an unsafe act.

Ground Rules for using HFACS

- Answer the HFACS questions in each bin. The questions will guide you through choosing the most appropriate codes for the unsafe acts and SIs.

- Avoid personal bias. Let the analyzed evidence guide you to the appropriate code. If one is attempting to be creative and make a code fit (calling a horse a zebra), then it is not applicable to explaining the cause-and-effect relationships within the mishap.

- Every unsafe act will include at least one individual system inadequacy. There are generally multiple reasons why an individual failed. Whether the mishap person's unsafe action was the result of leader/supervisory, training, support, or standards influences or not, each individual unsafe act will have at least one individual SI.

- Think cause and effect. SIs may apply directly to the individual's act or indirectly through other SI codes that influenced the unsafe act. Without a ***thorough analysis of evidence***, the safety officer, safety specialist, or SIB can easily lose sight as to how an SI code is applicable to the individual's unsafe act or the individual SI.

- Avoid rabbit holes. If a code becomes debatable, be willing to move on and come back. It is very easy to become distracted from the facts by debating whether a specific code applies. Do not be afraid to consult with USACRC G-3 Investigations personnel or human factors experts (i.e., psychologists or physiologists) when you or your team require clarity about "why" something did or did not happen.

- Be willing to eliminate codes. Some codes may seem similar, yet you will find that certain codes are more applicable than others. Be willing to deselect codes that are contentious. Focus on those codes that best support the identified causal factors and will best support recommendations for corrective actions.

- There are no minimums or limitations on the number of system inadequacies selected. Although there is only one unsafe act code per finding, there may be numerous SIs. Do not feel pressured to select a specific number of codes. However, each code requires written analysis related to the causal factors in the selection of the code. When in doubt, always refer to the evidence.

- Think about which codes best support needed corrective actions. For every code selected, the safety officer/specialist or SIB should consider the level of importance the code has in resolving SIs within the unit, the command, or the larger organization. The goal of the HFACS is to improve the organization versus an individual.

- When in doubt, follow the evidence. If you struggle linking an SI to the mishap person's act or how the SI affected the severity of damage or injury, always refer to the "cause and effect" analysis.

Unsafe Acts

Unsafe acts are factors that are *"most closely tied to the mishap and can be described as unsafe act(s) or actions committed by the operator (mishap person) that result in human error or unsafe situation."*

There are two distinct groups of unsafe acts: (1) Errors (Judgment / Performance) and (2) Known Deviations.

Was the unsafe act an error or known deviation?

Errors: Unknown and/or unintended deviations where the individual's mental or physical activities failed to achieve their intended outcome which resulted in the mishap. Some examples of an error include the following: attempted a task without needed assistance; improper use of personal protective equipment (PPE) or safety devices; misjudged changes in surrounding environment; misjudged time, speed, clearance, distance, degree or angle, center of gravity, or attitude; failed to effectively react to hazardous conditions, objects, or a situation; etc.

There are two sub-groups of errors:

(1) judgment and decision-making errors
(2) performance-based errors

Known Deviations: Known, intended and deliberate deviations of standards, rules, regulations, instructions, or procedures by the mishap operator/aviator/worker, yet the negative outcome (mishap) was unintended. Known deviations may be the result of faulty logic in assessing risk, systemic undisciplined behavior that supervisors/leaders may or may not condone, or individual indiscipline.

These conscious deviations include but are not limited to the following: willfully taking shortcuts; operating beyond established speeds and/or safe distances; choosing to not use required safety devices/PPE/restraints; operating beyond a safe degree of angle, center of gravity or attitude; etc

Did the mishap person(s) make a performance-based error?
(If yes – choose the most applicable error code. If no – go to the judgement-based error codes.)

Performance / Skill Based Errors. Occur when a specific action or inaction (control input, brake application, no inspection etc.) is performed in a manner that leads to mishap.

Unintended Activation or Deactivation: is a factor when an individual's movements inadvertently activate or deactivate equipment, controls, switches, weapons systems, etc., when there is no intent to do so. This action may be noticed or unnoticed by the individual at the time of occurrence. The error may be the result of one of more individual physical or mental conditions, crew/team influence, supervisory/leader influence, or a flaw in workspace or materiel design.	**AE101**
Procedure or Checklist Not Followed Correctly: is a factor when the mishap individual did not follow correct procedure. (Examples include failed to execute proper sequence, learned maneuver or proper emergency procedures; failed to follow a published checklist, Technical Manual (TM), or Standard Operating Procedure (SOP) to perform an inspection or maintenance of aircraft/vehicle/vessel/equipment, etc.) The error results from one or more individual system inadequacies, supervisory influence and/or ineffective training.	**AE102**

Over-Controlled / Under-Controlled Aircraft/Vehicle/Vessel or System: is a factor when the mishap individual's reactionary response was inappropriate for conditions by either over- or under-controlling the aircraft/vehicle/vessel/system. (Examples include too much or too little pressure, oversteering/understeering, improper braking, etc.) The error results from one or more individual system inadequacies, supervisory influence, and/or ineffective training.	**AE104**
Breakdown in Visual Scan or Instrument Cross-check: is a factor when the mishap individual did not effectively execute learned/practiced internal or external visual scan patterns. The error results from one or more individual system inadequacies, supervisory influence, and/or ineffective training.	**AE105**
Rushed or Delayed a Necessary Action: is a factor when the mishap individual takes the correct action(s) as dictated by the situation but performs the action(s) either too quickly or too slowly. (Examples include actuated a brake or device too soon or too late, etc.) The error results from one or more and/or supervisory influence and/or ineffective training.	**AE107**
Misinterpreted/Misread Instrument: is a factor when the individual is presented with a correct instrument reading, but its significance is not recognized, it is misread or is misinterpreted. This may be associated with ineffective training, supervisory influences, environmental factors, or other individual Sis.	**AE108**

Unsafe Acts

Was a mishap person's act from poor judgment or decision-making? (If yes – choose the most applicable error code, then move onto identify the associated individual system inadequacies. If no – go to known deviations.)	
Judgment and Decision-Making Errors: Exist when the individual pursues an inappropriate course of action after intentionally or unintentionally failing to accurately assess a situation which resulted in a mishap. The error is an unknown deviation of a policy or procedure during the performance of diagnostic or problem-solving tasks that require conscious effort. These may be the result of individual conditions, crew/team influence, leader/supervisor influence, ineffective training, and/or other organizational influences.	
Inadequate Real-Time Risk Assessment / Action: is a factor when an individual failed to adequately evaluate the risks associated with a particular course of action during execution of a task/mission/activity, and this faulty evaluation leads to inappropriate decision-making. (Examples include made an incorrect decision regarding immediate hazardous conditions; omitted use of PPE or safety devices, etc.)	**AE201**
Ineffective Task Prioritization: is a factor when the mishap individual failed to effectively organize and accomplish the tasks required to manage a situation.	**AE202**
Ignored a Caution/Warning: is a factor when the mishap individual disregarded an accurately perceived and understood caution or warning in favor of addressing what they perceive to be a greater immediate threat. (Examples include a sign, signal, guard, audible alarm, flashing light, verbal communication, etc.) The error may be a result of competing inputs/priorities, preconditions of the individual, the operating environment, crew/team influence, leader/supervisor influence or ineffective training.	**AE205**
Wrong Choice of Action During an Operation: is a factor when the individual, through faulty logic, inexperience, or erroneous expectations, selected the wrong course of action. (Examples include drove too fast for conditions; attempted task without needed or required assistance; used PPE or safety devices improperly; pulled or pushed improperly; mounted or dismounted a vehicle, equipment, obstacle, or platform improperly, etc.)	**AE206**
Misjudged / Misperceived Changing Environment: is a factor when an individual misperceived or misjudged altitude, separation, clearance, speed, closure rate, road or sea conditions, degree of angle, aircraft/vehicle location within the performance envelope or other operational conditions. (Operating a ground motor vehicle, boat, or aircraft during day or night, etc.) This may be the result of individual conditions, environmental factors, ineffective training, and/or supervisory/leader influences.	**AE207**

Did the mishap person commit a known deviation of policy or procedure?	
Known Deviations: Are known, intended and deliberate deviations of standards, rules, regulations, instructions, or procedures by the mishap operator/aviator/worker yet, the negative outcome (mishap) was unintended. Known deviations may be the result of faulty logic in assuming risk, systemic undisciplined behavior that may or may not be condoned by supervisors/leaders or solely individual indiscipline. These conscious deviations include but are not limited to the following: Willfully taking shortcuts; operating beyond established speeds and/or safe distances; choosing to not use required safety devices/PPE/restraints; operating beyond a safe degree of angle, center of gravity or attitude; etc.	
Performed Known Deviation (Work-Around): is a factor when the mishap individual disregarded published policy/guidance/procedure to pursue what he/she believed to be the best course of action based on available information to make a real-time risk decision. (Examples include chose to drive/operate outside published limits, by-pass safety procedures, not use available PPE, etc.) These deviations may have been well intended however, they disregarded established policies and safe work practices. The "shortcut" may be due to lack of resources (funding, personnel, tools, equipment, etc.), operational tempo (OPTEMPO), a lack of knowledge, or a lack of detail in guidance from supervisors/leaders. Work-around solutions and unofficial procedures that are accepted by leaders / supervisors within a community and considered necessary for certain operations are also captured under this code.	**AD001**
Commits Widespread/Routine Known Deviation (Normalization of Deviance): is a factor when the mishap individual violates published standard, procedure or policy based on unofficial accepted practices of the unit or community that are routine, ongoing or widespread. These chronic "bending" of the rules may or may not have leadership sanction yet have not routinely resulted in disciplinary/administrative action (culturally accepted) creating a normalization of deviance.	**AD002**
Extreme Lack of Discipline (Indiscipline): is a factor when an individual is trained to standard, knows the standard, but elected not to follow the standard without cause or need.	**AD003**
Finished with Unsafe Acts, proceed to the Human Error Flow Chart.	

Human Error Flow Chart

Note: Individual and Leader questions (highlighted by yellow background) are positive biased and prompt the reader to follow the 'yes' line to determine failure. Support, Training, and Standards questions are negative biased and prompt the reader to follow the 'no' line.

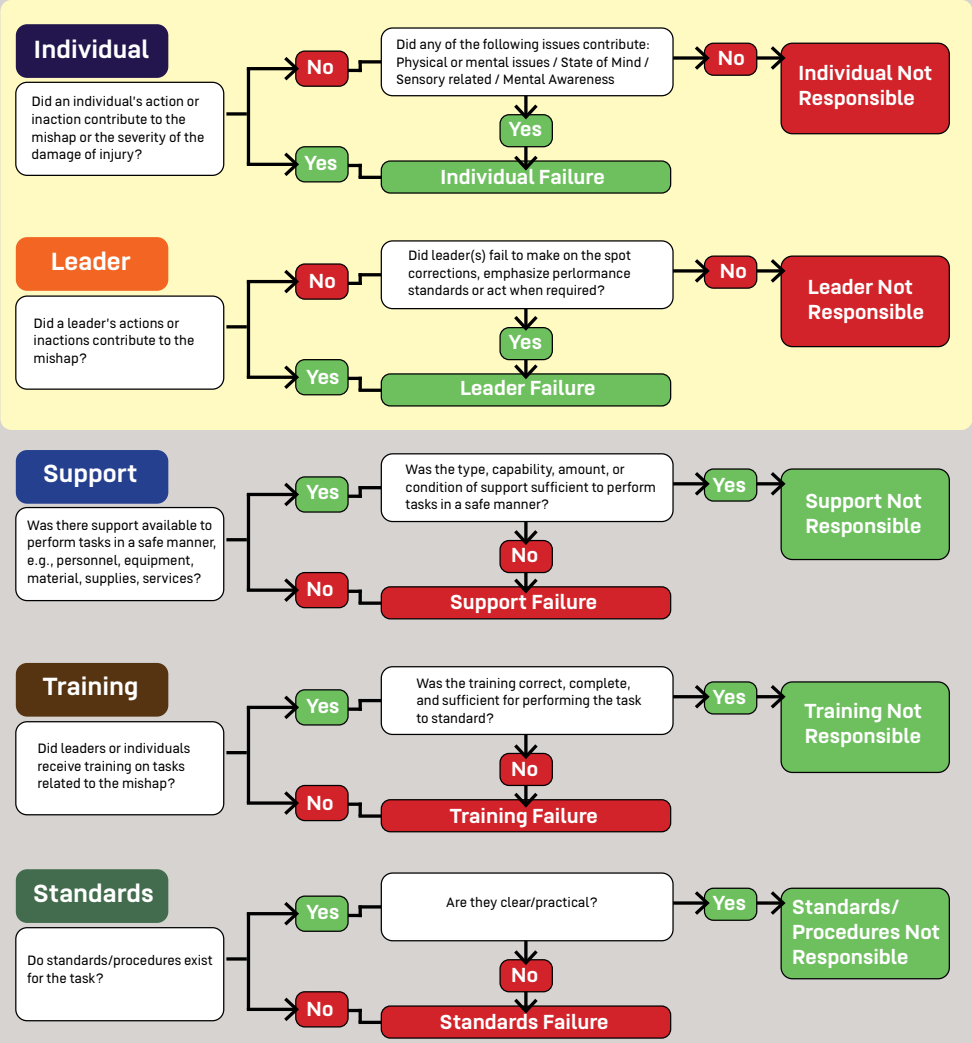

Figure 3: Human Factor Flow Chart

Individual System Inadequacies

Individual system inadequacies include the mishap individual's physical, mental, or cognitive conditions, and/or interactions with the physical environment and/or their equipment that negatively affected the performance or actions of the mishap person. Individual lifestyle behaviors, supervisor/leader influences, influences in training, resource support, policy/standards, or a combination thereof influence the actions of an individual. **Therefore, *at least one individual system inadequacy will accompany an unsafe act.***	
Question I1: Did a mental awareness condition of the mishap individual influence the unsafe act? *(If yes, determine which code(s) influenced the unsafe act. If no, skip to I2.)*	
Mental Awareness Conditions: Exist when the mishap individual experienced a failure in attention management which negatively affected the mishap individual's perception and/or performance and resulted in a hazardous condition or unsafe act. ***NOTE: It is best to consult with a qualified medical professional regarding these codes.***	
Inattention: is when the mishap individual did not maintain a state of readiness or alertness/situational awareness to properly act upon available information and resulted in a hazardous condition or unsafe act. This may have been due to boredom, self-confidence, over-reliance on automation, high experience levels, executing highly repetitive tasks where the mishap individual was on "auto-pilot," or a false sense of security or a perceived absence of threat from the environment (sheer lack of attention/awareness of risk). ***NOTE:* This code rarely is selected as a stand-alone code. Consider additional individual or leader code (s) as well.**	**PC101**
Fixation (Channelized Attention): is when the individual is focused all conscious attention on a limited number of environmental cues to the exclusion of others which resulted in a hazardous condition or unsafe act. This may be described as a tight focus of attention that led to the exclusion of comprehensive situational information. ***NOTE: If the fixation was the result of a task saturation, then use PC103.***	**PC102**
Task Saturation: is when the quantity of information an individual was processing exceeded their mental resources in the amount of time available and resulted in a hazardous condition or unsafe act. In other words, there is simply too much to accomplish with not enough time, not enough resources, and far too many tasks and tactics to accomplish simultaneously. This could be real or imagined but can result in both performance and judgment and decision-making errors.	**PC103**
Confusion: is when the mishap individual was unable to maintain a cohesive and orderly awareness of events and required actions and experienced a state characterized by a lack of understating, clear thinking or sometimes a misperception of the situation which resulted in the hazardous condition or unsafe act.	**PC104**
Negative Habit Transfer: is when the individual reverts to a highly learned behavior used in a previous system or situation and that response is inappropriate for current task demands and resulted in a hazardous condition or in unsafe act. (Examples include transitioned to a new piece of equipment, vehicle, vessel, or aircraft and using the same process habits to operate the new system which is incorrect.)	**PC105**

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Individual System Inadequacies

Distraction / Interruption: is when the individual had an interruption of attention or inappropriate redirection of attention by either an environmental cue, technology, a mental process, or other human influence which resulted in a hazardous condition or unsafe act. This may include a momentary interruption which resulted a subsequent failure to complete the original task or resulted in skipping steps in the original task.	**PC106**
Geographically Lost: is when the individual is at a different location from where one believes they are.	**PC107**
Change Blindness/Inaccurate Expectation: is when an individual's expectations contributed to not perceiving the change or to false interpretation of perceived stimuli. The stimulus would be easily noticed by the individual if he/she were directed to the change/reality. This is a universal limitation of human attention. ***NOTE: This code should be used in place of PC101 (inattention) if the safety investigator believes the lack of attention involved a limitation on the ability to perceive the stimulus/change.***	**PC110**

Question I2: Did the mishap person(s) state of mind influence the unsafe act? (If yes, determine which code(s) influenced the unsafe act. If no, skip to I3.)	
State of Mind Conditions: Exist when psychosocial problems, life stressors, personality traits, or misplaced motivation of the mishap individual created a hazardous condition or unsafe acts. Consult with a medical professional when assessing these codes.	
Psychological Disorder: is when the individual has met criteria for a diagnosable psychiatric disorder by a competent medical professional which, in their opinion, resulted in a hazardous condition or unsafe act. This may or may not have been diagnosed prior to the mishap.	**PC202**
Life Stressors / Emotional State: is when the individual's emotional state and/or life circumstances lead to burnout or otherwise degrade performance which resulted in a hazardous condition or unsafe act. This can occur when one feels overwhelmed, emotionally drained, and/or unable to meet constant demands which impinge on performance. (Examples include wearying effects of work, training, relationships, economic or legal stressors, housing difficulties, upcoming/recent change of station, new baby, family or personal medical issues, or a combination of circumstances, etc.) ***NOTE 1: This may be associated with PC307- Fatigue and/or other individual SIs.*** ***NOTE 2:* This code rarely is selected as a stand-alone code. Consider additional leader code (s) as well.**	**PC203**
Personality Style: is when evidence strongly indicates the individual's personal interaction with other personalities created a hazardous condition or unsafe act. Examples range from over-conservative, authoritarian, overly aggressive, persuasive, impulsive, invulnerable, submissive, passive/non-assertive, or other personality traits that result in degraded performance. For this code to be selected, there must be strong evidence that the individual's personality traits are longstanding and pervasive and fall outside of the "reasonable person" concept for similar individuals doing similar tasks/missions. **NOTE: This code rarely is selected as a stand-alone code. Consider additional leader code (s) as well.**	**PC205**

Overconfidence: is when the individual unreasonably overvalues or overestimates their own capability, the capability of others or the capability of aircraft/vehicle/vessel or equipment. For this to be selected, there must be strong evidence the individual acted in a manner inconsistent with the "reasonable person concept" (this individual's overestimation is above and beyond what a reasonable person in a similar situation would have been expected to do). **NOTE: This code rarely is selected as a stand-alone code. Consider additional leader code (s) as well.**	**PC206**
Pressing, Haste, Motivation: is when the individual's motivation to complete a task/mission was misplaced, and/or the individual knowingly pressed their self and/or equipment beyond reasonable capabilities which resulted in a hazardous condition or unsafe act. This also includes excessive or weak motivation when either the weak or excessive motivation to succeed disregarded mission needs or superseded the goals of the unit (e.g., rushed to complete a task, mission, or reach a destination; or demonstrated a weak/excessive motivation that increased risk to self and/or team). This SI may be the result of internal or external pressures or influences on the individual. ***NOTE:* This code rarely is selected as a stand-alone code. Consider additional leader code (s) as well.**	**PC209**

Question I3: Did the mishap person have a medical or physical condition that negatively affected performance and influenced the unsafe act? (If yes, determine which code(s) influenced the unsafe act. If no, skip to I4.)	
Adverse Physiological Conditions: Exist when an individual experiences a physiologic event that compromised performance resulting in a hazardous condition or unsafe acts. ***NOTE: When using any of the PC300 codes, ensure you consider consulting with qualified professionals who specialize in the areas of physiology, as defined by your service.***	
Effects of Gravitational Forces (G-LOC): is when, in aviation, the individual experienced G-induced loss of consciousness (G-LOC), grey-out, blackout or other neuro-circulatory effects of sustained acceleration forces.	**PC301**
Substance Effects (supplements, medications, drugs, alcohol): is when the use of authorized or unauthorized substances (medications, supplements, energy enhancing products, alcohol, illegal drugs, etc.) negatively affected performance and resulted in a hazardous condition or unsafe act.	**PC302**
Loss of Consciousness (sudden or prolonged onset): is when the individual experienced a loss of consciousness/functional capacity for a few seconds or prolonged and resulted in degraded performance. Causes include low oxygen atmosphere, trauma, shallow water blackout, or any other cause resulting from activities that were directed, supervised or self-initiated. (NOT GLOC - see PC301.)	**PC304**

Continued on next page

Individual System Inadequacies

Physical Illness/Injury: is when pre-existing or operationally related medical conditions (illness, injury, dehydration, motion sickness, trauma, seizure, toxic chemical exposure, etc.) negatively affected performance and resulted in a hazardous condition or unsafe act. ***NOTE: Do not use this code to capture injury or illness that does not cause a hazardous condition or an unsafe act.***	**PC305**
Physical Overexertion: is when the individual's diminished physical capability caused by overuse (time/relative load) resulted in a hazardous condition or unsafe act. The effects of prolonged physical activity, or the effects of brief but relatively extreme physical activity, either of which depletes a person's physical endurance or strength beyond the individual's normal limits and degrades performance.	**PC306**
Fatigue: is when acute or chronic sleep deprivation or circadian rhythm disruption (shift work/lag, extended duty periods, jet lag, poor sleeping conditions, etc.) negatively affected physical and/or mental performance and resulted in a hazardous condition or unsafe act. ***NOTE: Fatigue should be quantified by determining the mishap individual's number of hours awake vs sleep and activity in the past 72 hours leading up to the mishap. Also determine if the fatigue was either self-induced or operationally induced.***	**PC307**
Acute Trapped Gas Disorders: is when gases in the middle ear, sinuses, teeth, or gastrointestinal system negatively affected performance and resulted in a hazardous condition or unsafe act. (Occurs in aviation or diving.) ***NOTE: If the alternobaric vertigo (dizziness from unequal pressures between each ear) induced spatial disorientation, you must also include PC321.***	**PC310**
Decompression Sickness (Evolved Gas Disorder): is when evolved gases negatively affected performance resulting in hazardous condition or unsafe act. *(Examples include bends, chokes, central nervous system manifestations, paresthesia, etc.)*	**PC311**
Respiratory Physiological Event: is when hindered/inappropriate respiration or pressure/flow/concentration of oxygen created respiratory physiological symptoms which negatively affected performance resulting in a hazardous condition or unsafe act.This can be caused by external forces and/or internal metabolic functions. Use this code to capture symptoms developed due to hyper/hypoxia, hyper/hypoventilation, and other metabolic conditions which result in a disruption of metabolic balance.	**PC312**

Inadequate Adaptation to Darkness: is when dark adaptation was either not fully completed or light washed out night vision and negatively affected the individual's performance resulting in a hazardous condition or unsafe act. ***NOTE 1:*** **This code rarely is selected as a stand-alone code. Consider additional individual or leader code (s) as well.** ***NOTE 2: If the ineffective adaptation to darkness was related to lights of a vehicle/vessel/aircraft, consider applicability of PE109 in addition to this code.***	**PC314**
Anthropometric/Biomechanical Limitations: is when the individual's size, strength, dexterity, coordination, endurance, or other physical factors negatively affected performance which resulted in a hazardous condition or unsafe act.	**PC317**
Nutrition/Diet: is when evidence supports the individual's nutritional state negatively affected performance.	**PC319**
Loss of Capacity (Surprise/Startle Response): is when uncontrollable, automatic physical response of muscle reflex, raised heart rate, suddenly dropped blood pressure, etc. was elicited by exposure to a sudden, intense event that contradicts an individual's expectations resulting in a hazardous condition or unsafe act. This response can affect the physical and mental processes normally used to effectively respond to the event/emergency.	**PC320**
Spatial Disorientation: is when the individual failed to correctly sense a position, motion, or attitude of the aircraft or his/herself within the fixed coordinate system provided by the surface of the earth and the gravitational vertical position (e.g., visual, vestibular, kinesthetic, or auditory/sound illusions) which resulted in a misjudgment and unsafe act. ***NOTE 1: This may be the result of other individual preconditions, ineffective training, and/or supervisory/leader influences.*** ***NOTE 2: Do not use this code if the disorientation is the result of substance effects; use PC302.***	**PC321**

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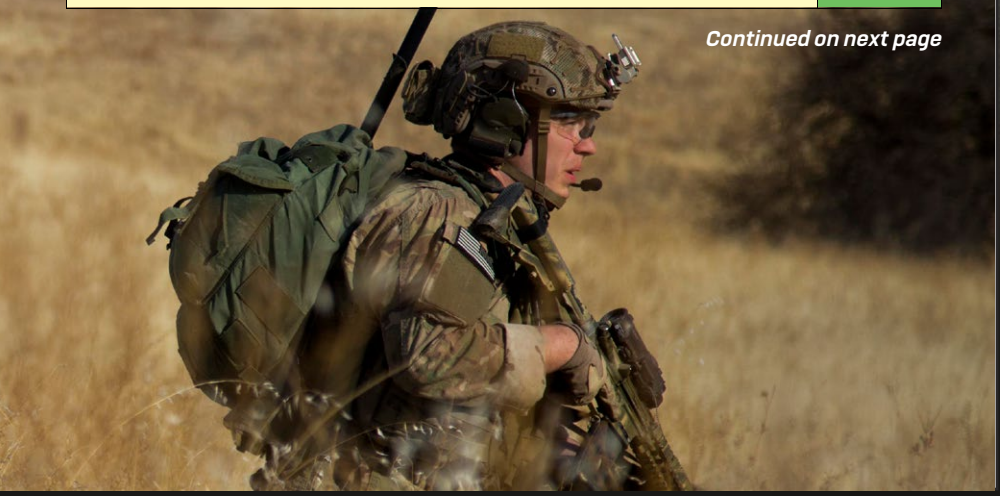

Individual System Inadequacies

Question I4: Did conditions of the operational physical environment affect the actions of the mishap individual? (If yes, determine which code(s) influenced the unsafe act. If no, finished with individual system inadequacies.)	
Physical Environment: Are conditions related to the immediate physical surroundings which negatively affected human performance resulting in unsafe acts. These are conditions such as terrain surfaces, physical obstructions, noise, illumination, glare, air contaminants (e.g., gases, fumes, vapors, particulates), low oxygen, vibrations, radiation, wildlife, insects, and meteorological conditions (e.g., precipitation, temperature, humidity, pressure, wind, electromagnetic effects, and lightning). These codes do not typically "stand alone" requiring the SIB to also consider other system inadequacies.	
Environmental Conditions Affected Vision: is when conditions such as lighting/illumination, physical obstructions, rain, snow, spray, fog, haze, darkness, smoke, dust, sand, other particulates, etc., impeded clear viewing/vision and negatively affected performance and resulted in hazardous conditions or unsafe acts.	**PE101**
Vibration Affected Performance: is when the intensity and/or duration of vibrations from an engine, equipment, tools, airframe, rotor, and/or propeller negatively affected vision, balance, and/or performance and resulted in hazardous conditions or unsafe acts.	**PE103**
Temperature Affected Performance: is when the ambient/workspace temperature negatively affected performance and resulted in hazardous conditions or unsafe act. ***NOTE: If this code is selected, consider if PC305 is applicable.***	**PE106**
External Force or Object Impeded Performance: is when accelerative forces, wind, sea-state, objects, aircraft/vehicle/vessel structures, etc. impeded individual movement and resulted in hazardous conditions or unsafe acts. ***NOTE: Use PC304 to code GLOC.***	**PE108**
External Lighting of Vehicle/Aircraft/Vessel/Object Affected Vision: is when the intensity, position, pattern, color, and/or absence of the lighting of other aircraft, vehicles, vessels, or object negatively affected performance and resulted in hazardous conditions or unsafe acts. ***NOTE:* This code rarely is selected as a stand-alone code. Consider additional individual or leader code (s) as well.**	**PE109**
Noise Interference: is when an unexpected sound (bang, explosion, shout, alarm, machine noise, etc.) not directly related to information needed for a task negatively affected performance and resulted in hazardous conditions or unsafe acts.	**PE110**
Terrain Feature Affected Performance: is when known yet unanticipated or unseen/unknown terrain hazards were encountered which negatively affected performance and resulted in hazardous conditions or unsafe acts. (Uneven surfaces, loose gravel or rock, sand, soft shoulders, pooled water, black-ice, pothole, drop-off, dense vegetation, deep mud, etc.)	**PE112**
Return to Human Error Flow Chart	

Leader System Inadequacies

Leader System Inadequacies are longstanding actions or inactions, methods, or directives of any supervisory/leadership personnel within the unit that created hazardous practices or conditions and resulted in unsafe acts. This includes any personnel serving in a leadership/supervisory role ranging from the immediate supervisor to the commanding officer/director. (E.g., VC/TC, shift supervisor, pilot in command, squad leader, platoon leader, team leader, 1SG, etc.) Each position may have influenced subordinate actions or behaviors in the months, weeks, days and/or minutes leading up to the individual's unsafe act. When choosing codes applicable to these system inadequacies, the SIB must put into context of where the leader errors or known deviations reside (first-line leader/supervisor, etc.). If leadership flaws include multiple first line and/or second line supervisors and/or command leadership, ensure selected codes address each level of error.	
Question L1: Were the actions of the leader inadequate? (If yes, determine which code(s) influenced the unsafe act or other system inadequacies. If no, skip to L2.)	
Ineffective Supervision: Exist when supervisory/leadership personnel failed to identify and assess hazards, mitigate risks, train, and brief personnel, and/or provide effective guidance and oversight which resulted in hazardous conditions or unsafe acts.	
Ineffective Supervisory or Command Oversight: is when the availability, competency, quality, or timeliness of supervisor/leader oversight did not meet task or mission demands which resulted in hazardous conditions or unsafe acts. (Examples include failure to verify accuracy and completeness of work, conduct pre-combat checks / pre-mission inspections, mismanagement of emerging risks during mission execution, etc.) ***NOTE: This code also includes Inappropriate supervisory pressures.***	**SI001**
Lack of Supervisory Responses to Critical Information: is when an individual or crew provided critical information regarding a potential safety issue of the operating environment, equipment, or personnel to supervisory/leadership personnel, who failed to act/close the loop, which resulted in a hazardous conditions or unsafe acts.	**SI006**
Failed to Identify or Correct Hazardous Practices, Conditions or Guidance: is when any supervisor/leader in the unit failed to identify or correct known hazardous conditions of equipment, facilities, or written procedures/guidance, or correct unsafe work practices of personnel within his/her scope which resulted in hazardous conditions and/or unsafe acts.	**SI007**
Tasked Individual(s) with Lack of Experience, Currency or Proficiency: is when a supervisor/leader inadvertently tasked an individual or team whose fluency or expertise did not match skills required for safe execution of the task, system, or mission; or whose familiarity with a task or process was either not current or limited by infrequent or rare performance and resulted in hazardous conditions or unsafe acts. This may be due to flaws in institutional or local training or a leader's lack of knowledge of his/her personnel. ***NOTE: If the supervisor was aware the individual or team's skill levels were inappropriate for the task, then refer to SD003.***	**SI008**
Rank/Position Intimidation: is when a supervisor/leader caused the task performance capabilities to be degraded. Also, conditions where formal or informal authority gradient is too steep or too flat across a crew/team and this condition degrades collective or individual performance.	**SI009**

Continued on next page

Leader System Inadequacies

Unit OPTEMPO/Workload: is when the pace of workload, additional duties, training, deployments, or other workload-inducing conditions of a unit created hazardous conditions or unsafe acts.	**SC102**
OPTEMPO: is when the workload-inducing conditions set at brigade (BDE) or higher level negatively affected the ability of unit leaders to effectively manage risks which resulted in hazardous conditions and/or unsafe acts.	**OP001**

Question L2: Did a supervisor / leader knowingly deviate from a known rule, regulation, policy? (If yes, determine which code(s) influenced the unsafe act, or other system inadequacies, or the severity of damage or injury. If no, skip to L3.)	
Supervisory Known Deviation: Exists when a supervisor/leader willfully disregarded instructions, guidance, policies, rules, or standard operating procedures. This includes failing to enforce standards, allowing unwritten practices to become standard, directing individuals to violate existing rules / regulations and authorizing unqualified personnel for a task.	
Failure to Enforce Published Rules/Guidance: is when a supervisor/leader failed to ensure that personnel adhered to published rules/policy/guidance/procedure or knowingly allowed an untrained, inexperienced, non-proficient or non-current individual to perform a task which resulted in hazardous conditions or unsafe acts. (Examples include a failure to enforce a policy, standard operating procedures, or technical guidance in regard to vehicle/watercraft operation, weapons or explosives handling, use of machines or hand tools, etc.; failed to enforce use of PPE such as restraints, eye, face or head protection, hearing protection devices, etc.)	**SD001**
Allowed Unwritten Practices to Become Standard (Normalization of Deviance): is when a supervisor/leader chronically condoned the use of unwritten/unofficial procedures by subordinates which resulted in hazardous conditions or unsafe acts.	**SD002**
Directed Individual to Violate Existing Regulations, Rules, or Procedures: is when a leader/supervisor directed a subordinate to circumvent existing standard operating procedures, regulations, instructions, policies, or technical guidance which resulted in a hazardous condition or unsafe acts. This may be the result of the supervisor's personality, a faulty logic or when the consequences/risk of violating published procedures was recognized and determined by the supervisor/leader to be the best course of action.	**SD003**

Question L3: Was the pre-mission planning ineffective? Did leaders/supervisors fail to execute or implement the deliberate risk management process? (If yes, determine which code(s) influenced the unsafe act, or other system inadequacies, or the severity of damage or injury. If no, skip to L4.)	
Ineffective Planning and Coordination: Exist when unit leadership failed to effectively utilize the troop leading procedures/risk management process to assess hazards and develop effective controls associated with an activity, event, mission, or operation which resulted in unnecessary risk. Occasionally, leaders influence OPTEMPO or schedules creating unacceptable risk, jeopardizing team/crew rest, and adversely impacting performance. Such planning flaws, though arguably unavoidable during emergency or combat situations, are not acceptable for readiness training or day to day operations. Planning includes the risk management process from information collection, consideration of team member knowledge, skills and physical conditions, analysis of environmental hazards, resource support factors, equipment conditions, capabilities of external agencies, and contingency planning. Included in this category are issues of crew/team composition, pre-mission deliberate risk assessments, risk acceptance authority, information resource support, and personnel manning.	
Ineffective Deliberate Risk Assessment: is when supervision/leadership did not effectively apply the DoD risk management (identify hazards, assess hazards, develop controls, implement controls, supervise, and evaluate/assess) during pre-mission/activity/event planning or a job hazard analysis (JHA) which resulted in hazardous conditions and/or unsafe acts. This includes assessment of all hazards including crew/team composition. (Examples include did not have enough trained, licensed, certified, or qualified personnel to safely operate the number of vehicles or equipment available, or not enough personnel with specific occupational specialties required for the task or mission.)	**SP006**
Authorized Unnecessary Risk: is when a leader with risk acceptance authority, unnecessarily authorized a mission, activity, or task which resulted in hazardous conditions and/or unsafe acts. **NOTE: This code could be considered when the safety investigator concluded the residual risk was accepted at the wrong level as defined by services policies.**	**SP007**
Ineffective Pre-Mission Planning: is when supervision/leadership did not conduct effective pre-mission/activity/event planning which resulted in hazardous conditions and/or unsafe acts. **NOTE: This is outside of the formal/deliberate risk management process. If there were deficiencies in the risk management process or the deliberate risk assessment, then cite SP006.**	**SP008**

Continued on next page

Leader System Inadequacies

Question L4: Did the safety culture contribute to an unsafe act? (If yes, determine which code(s) influenced the unsafe act, or other system inadequacies, or the severity of damage or injury. If no, skip to L5)	
Safety Culture: occurs when the unspoken or unofficial rules, values, attitudes, beliefs, and customs of small unit leaders or their higher organization negatively affected good order and discipline in adherence to established safety standards, initiatives, and practices. A response of "that is the way things really get done around here" is an indicator. Other issues related to culture include unit justice, psychological contracts, and esprit de corps. All these issues affect attitudes and the value of a safe working environment. Leaders influence culture and different levels of leadership from the section or platoon up to the highest levels of command. Leaders can create a culture of cohesion in which individuals have a sense they belong, that they are valued, and that they commit to personal and organizational goals either in a safe manner or overly risky manner. Leaders in small units can also create a culture of high risk and distrust from overemphasis on time versus safe completion of tasks, lack of accountability, training, communication, consistent discipline, etc.	
Unit Culture (attitude/actions) Created Increased Risk: is when the explicit or implicit actions, statements, attitudes, techniques, or values of supervisors/leaders facilitated an environment where demands or pressures existed that resulted in hazardous conditions or unsafe acts. (Examples include mission or time expectations supersedes mishap prevention strategies; role-modeling fosters unsafe behaviors, proactive safety initiatives receive minimal support; near-misses, mishaps or adverse events are minimized or considered non-reportable; limited positive recognition for safe performance or reporting hazardous conditions; lessons learned are not shared throughout the entire unit; etc.)This also includes ineffective facilitation of a constructive climate such as establishing and maintaining an accurate and shared understanding of the evolving task or mission on the part of all personnel. ***NOTE: When using this code, consider applicability of SD002 and PP101.***	**SC101**
Organizational Culture (attitude/actions) Created Increased Risk: is when explicit or implicit actions, statements, attitudes, or techniques at a higher organizational level facilitated an environment where demands or pressures existed, resulting in hazardous conditions or unsafe acts throughout subordinate units or the field. OC 001 is applicable at the BDE and higher level.	**OC001**
Organizational Perceptions of Materiel Resources (Equipment): is when there is organizational over- or under-confidence in vehicle systems, vessels, aircraft, weapons systems, communication systems or any other materiel resulting in a hazardous condition or unsafe acts throughout subordinate units or the field / fleet.	**OC003**
Mission/Aircraft/Vehicle/Ship/Equipment Change or Unit Deactivation: is when the process of changing missions, aircraft/vehicle/ship/equipment or an impending deactivation resulting in hazardous conditions or unsafe acts throughout subordinate units or the field.	**OC004**
Organizational Structure is Unclear or Inadequate: is when the chain of command of subordinate commander(s) or structure of an organization is confusing, non-standard, or inadequate resulting in hazardous conditions or unsafe acts throughout subordinate units or the field / fleet. ***NOTE: This is applicable when subordinate leaders are receiving direction from more than one command.***	**OC005**

Question L5: Did challenges with teamwork contribute to an unsafe act? (If yes, determine which code(s) influenced the unsafe act, or other system inadequacies, or the severity of damage or injury. If no, finished with Leader Failures.)	
Team Coordination/Communication Conditions: refers to verbal or non-verbal interactions among crews/teams involved with the preparation and/or execution of a task/mission which resulted in hazardous conditions or unsafe acts. This includes failures with communication between members of aircraft, tactical vehicles, ground guides, boat or ship, stevedore/long shoring, or any other crew/team communication failures.	
Ineffective Team Resource Management (Crew, Bridge, Fighter, Maintenance, etc.): is when crew/team members failed to actively maintain an accurate and shared understanding of the evolving task or manage their distribution of tasks which resulted in a hazardous condition or unsafe act. This includes communication breakdowns (e.g., standardized terms, phrases, hand signals or language/lexicon barriers), critical information not shared, rank/position intimidation, lack of assertiveness or other teamwork functions.	**PP101**
Task/Mission Planning and/or Briefing Inadequate: is when an individual, crew or team failed to complete all preparatory tasks associated with planning the mission and/or effectively briefing the tasks which resulted in a hazardous condition or unsafe act. Planning tasks include information collection and analysis, coordinating activities within the crew or team and with appropriate external agencies, risk assessment followed by the pre-mission/task safety briefing. ***NOTE: You will need to review SP100 codes to determine if the individual's or team's failure was secondary to higher level planning and briefing flaws.***	**PP109**
Task/Mission-In-Progress Re-Planning: is when crew or team members fail to adequately reassess changes in their dynamic environment during mission execution and change their mission plan accordingly to ensure adequate management of risk which resulted in a hazardous condition or unsafe act.	**PP111**
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Support System Inadequacies

Support System Inadequacies are when insufficient resources are a factor. Specifically, when the type, amount, capabilities, or condition of the support is insufficient to correctly perform the mission or task. Support includes personnel, equipment, materiel, supplies, infrastructure services or facilities.	
Question SU1: Did an issue with resources create or lead to an unsafe situation or unsafe act? (If yes, determine which code(s) influenced the unsafe act, or other system inadequacies, or the severity of damage or injury. If no, skip to SU2.)	
Resource Support Problems: are system inadequacies when resource support or system safety shortfalls resulted in ineffective risk management or created hazardous conditions for leaders/supervisors and/or the operator/aviator/worker. Resource support problems exist when the type, amount, capabilities, condition of the resource support is not sufficient to correctly perform a mission. Resources include personnel, equipment, materiel, supplies, services, and/or facilities.	
Unit Failure to Provide Sufficient Operational Information Resources: is when unit/ship or installation level information resources (e.g., maps, graphic depictions, tables, charts, blueprints, weather data, intelligence, traffic, road, or terrain conditions etc.) were not made fully available to personnel who executed the mission/task/event and resulted in hazardous conditions and/or unsafe acts. ***NOTE: If the support failure was outside the unit, see OR008.***	**SP009**
Unit Failure to Provide Sufficient Manning/Staffing: is when unit/ship or installation planning processes failed to meet staffing demands or continuity of operations for ongoing missions and created an unnecessary OPTEMPO which resulted in hazardous conditions and/or unsafe acts. This may be the result of competencies, processes, internal policies, higher organizational manning support issues, etc. (Examples include reduced manning for holidays, contingencies, deployment manning needs, medical asset requirements to support training or events, etc.) ***NOTE: If the support failure was outside the unit, see OR007.***	**SP010**
Unit Failure to Provide Sufficient Equipment or Supplies: is when unit/ship or installation level leaders failed to ensure personnel executing the mission received all necessary equipment and/or supplies to effectively implement risk control measures which resulted in hazardous conditions and/or unsafe acts. (Examples include unnecessarily using not mission capable equipment, not providing personnel with necessary tools, communications equipment, kits, vehicles, PPE, etc.) ***NOTE: If the support failure was outside the unit, see OR004.***	**SP011**
Command and Control (C2) Resources are Deficient: is when installation or resources of command staff, maritime or airfield services, communications/IT support, etc. are unavailable or inadequate for safe operations resulting in hazardous conditions or unsafe acts throughout subordinate units or the field / fleet (e.g., Joint force or individual component C2, battlegroup management).	**OR001**

Inadequate Infrastructure: is when the support, maintenance and/or space provided by installations or industrial facilities supporting military operations or military production programs become inadequate, degraded, or non-available resulting in hazardous conditions or unsafe acts. This includes those facilities or services dedicated to deployment, reception, staging, movement, integration and sustainment, airfield services, dining, physical fitness, living quarters, recreation areas, petroleum, oil, and lubricant (POL) services, housing, medical clinics/hospitals, weather services, storage areas, maintenance facilities, property disposal, hobby shops, road maintenance, traffic management. ***NOTE: Traffic management includes road signs or overt identification of hazardous areas due to maintenance or environmental issues, etc.***	**OR003**
Purchasing or Providing Poorly Designed or Unsuitable Equipment: is when there are inadequacies in the acquisition and/or fielding of warfighting or commercial materiel resulting in hazardous conditions or fallible decisions throughout subordinate units or the field / fleet. *(Examples include failure to field needed equipment/materiel in a timely manner, failure to installation, inspections, quality control, or depot level maintenance services in a timely manner. Includes a breakdown in capability-based assessments, DOTMLPF-P change recommendations (DCRs), capability development documents (CDDs) or capability production documents (CPDs)).* ***NOTE: This code is likely a latent failure to "Technological Environment" findings.***	**OR004**
Failure to Remove Inadequate/Worn-Out Equipment in a Timely Manner: is when the process through which equipment is removed from service is inadequate resulting in hazardous conditions or unsafe acts throughout subordinate units or the field / fleet (Examples include tactical vehicle systems or components, shipboard systems or components, aircraft, etc.)	**OR005**
Personnel Accession/Selection Policies or Processes: is when the process through services recruit, screen, brought into the service, or placed into occupational specialties individuals is ineffective resulting in hazardous conditions or unsafe acts throughout subordinate units or the field / fleet. This occurs prior to occupational specialty training and includes screening processes for disqualifying conditions such as a physical, psychological, or behavioral disorders, aptitude, or history of any condition that may reasonably be expected to interfere with the successful performance of duty or training or limit geographical assignment.	**OR006**
Failure to Provide Adequate Personnel/Staffing Resources: is when the process through which personnel resource allocations, staffing or personnel placement processes are inadequate for mission demands resulting in hazardous conditions or unsafe acts throughout subordinate units or the field / fleet.	**OR007**

Continued on next page

Support System Inadequacies

Failure to Provide Adequate Information Resources: is when weather, intelligence, operational planning material or other information necessary for safe operations planning are too complex, too vague, incorrect, or not available throughout the organization resulting in hazardous conditions or unsafe acts throughout subordinate units or the field / fleet. This also includes knowledge management tools or data collection and analysis tools to support large safety management system programs such as materiel management, systems safety, hazard inspections and assessments, risk management, etc.	**OR008**
Failure to Provide Adequate Funding: is when subordinate organizations or an operation does not receive the financial resources to complete its assigned mission/task resulting in hazardous conditions or unsafe acts throughout subordinate units or the field / fleet. (Examples include leaders/supervisors are forced to find creative means to accomplish the mission which may create high risk conditions related to readiness training and/or use of materiel/equipment. Specific circumstance may force leaders to accept a higher level of risk in a specific area due to lack of funding for materiel, supplies, or personnel.)	**OR009**

Question SU2: Did the technological environment (workspace design) negatively affect the performance of the mishap individual or team? (If yes, determine which code(s) influenced the unsafe act, or other system inadequacies, or the severity of damage or injury. If no, finished with Support Inadequacies.)	
Technological Environment: is when workspace design conditions or automation affected the actions of individuals and resulted in a hazardous condition or unsafe acts. This includes ground vehicle systems, aircraft, watercraft / shipboard spaces, control stations, weapons systems, communication systems, maintenance repair systems, etc. ***NOTE: This section assesses hazardous conditions of materiel components and the role a materiel design condition or decision impacted an individual's actions.***	
Restraint System and/or Seat Problems: is when the design of a restraint system, seat, ejection system, and/or associated comfort elements impeded occupant performance or failed to function as intended which resulted in a hazardous condition or unsafe act.	**PE201**
Instrumentation and Warning System Issues: is when workspace/cockpit instrument or warning system elements (design, reliability, lighting/backlighting, audible cues, location, symbology, size, display, etc.) negatively affected performance which resulted in a hazardous condition or unsafe act. **NOTE: This also includes alarm fatigue and/or habituation.**	**PE202**
Workspace Visibility Restrictions (not weather related): is when obstructions from workspace design/layout prevent necessary visibility and negatively affected performance resulting in a hazardous condition or unsafe act. This includes physical design, glare, reflections, etc. ***NOTE: Visibility restrictions due to weather or other environmental conditions are captured under PE101.***	**PE203**
Controls and/or Switches: is when the location, shape, size, design, reliability, or other aspect of controls and/or switches negatively affected performance and resulted in a hazardous condition or unsafe act.	**PE204**
Automated System Created a Hazardous Condition: is when the design, function, reliability, symbology, logic, or other aspects of automated systems negatively affected performance which resulted in a hazardous condition or unsafe act. This includes designs of tooling machines, ship, or aircraft components, etc. ***NOTE: Use PE202 if alarm fatigue and/or habituation was the condition.***	**PE205**
Workspace Limitations Affected Performance: is when conditions of a workspace configuration/design that negatively affected performance which resulted in a hazardous condition or unsafe act.	**PE206**
Personal Equipment Interference: is when the individual's personal equipment negatively affected performance and resulted in a hazardous condition or unsafe act. This includes body armor, harness, other PPE, night vision devices (NVDs), weapons, etc.	**PE207**
Communication Equipment Ineffective: is when a communication system (voice, data, multi-sensory) limitation and/or malfunctions negatively affected performance and resulted in a hazardous condition or unsafe act.	**PE208**
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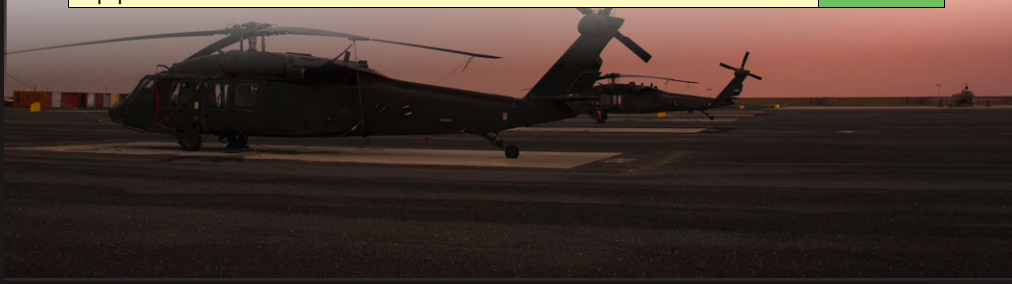
Training System Inadequacies

Training system inadequacies are when either unit level or institutional training was found to be incorrect, incomplete, or insufficient for performance?	
Question T1: Was any aspect of a training incorrect, incomplete, or insufficient for performance? (If yes, determine which code(s) influenced the unsafe act, or other system inadequacies, or the severity of damage or injury. If no, finished with training system inadequacies.)	
Individual Training Conditions: is when formal or informal instruction, skill development or knowledge limit the individual's capability, capacity or performance resulting in an unsafe act.	
Untrained Operator/Worker: is when the mishap individual did not receive adequate/sufficient training (formal, just-in-time, on the job) or received no training for a specific task which resulted in a hazardous condition or unsafe act. ***NOTE: Selection of this code requires assessment of various leader system inadequacy codes.***	**PT101**
Knowledge Retention: is when the mishap person did not remember information from training and/or previous experience necessary to complete a task safely which resulted in a hazardous condition or unsafe act. This may be due to flaws in local/unit training or a formal training program or the individual's capacity to learn and retain information. ***NOTE 1: Exposure to information at one point in the past does not imply "knowledge" of it.*** ***NOTE 2: Selection of this code requires assessment of various leader system inadequacy and/or training program codes.***	**PT102**
Lack of Currency: is when an individual's familiarity with a specific task or process is either not current or is limited by infrequent or rare performance of the task to permit safe execution which resulted in a hazardous condition or unsafe act. (e.g., The mishap individual was once trained to proficiency to operate a specific system or perform a process but has not done so in many months or years.) ***NOTE: Selection of this code requires assessment of appropriate leader system inadequacy (SI008, SP006, etc.)***	**PT103**
Lack of Proficiency/Experience: is when an individual's level of fluency or expertise does not match skills required for safe execution, regardless of one's familiarity with the process, task, system, or mission which resulted in a hazardous condition or unsafe act. ***NOTE: Selection of this code requires assessment of appropriate leader system inadequacy (SI008, SP006, etc.)***	**PT104**
Lack of Job/Work Related Safety Training: is when an individual has not received required or effective safety training related to hazards associated with their daily job or a one-time task, or when there is a change in process or equipment which resulted in a hazardous condition or unsafe act.	**PT105**



Question T2: Did established training programs contribute or result in an unsafe act?	
Training Programs: are when a training and/or educational program of instruction designed to improve technical, tactical, critical thinking or leadership skills is incorrect, incomplete, or insufficient for performance to standard which negatively influenced supervisor/leader and/or individual performance. ***NOTE: 1: The failure of an individual to absorb the training material in an adequate training program does not indicate a training program problem. (See PT102.)*** ***NOTE 2: The failure of an individual to recall learned information under stress or while fatigued despite attending an adequate training program does not indicate a training program problem. (See PC100, PC200, and/or PC300 series.)***	
Unit Failed to Provide Effective Training: is when supervisors/leaders failed to provide effective training to ensure competency and proficiency of their personnel's required skills or for a specific task which resulted in a hazardous conditions or unsafe acts. (Examples include technical procedures for amphibious assault/towing/convoy/flight, ground guiding, weapons handling, maintenance procedures, flight operations, working with hazardous materials, use of PPE, fall-protection, water survival, helicopter rope suspension techniques, combat tactics, techniques and procedures, firefighting, operation of specific variant/class of a vehicle/vessels or materiel handling equipment operation, etc.)	**SI003**
Resident Formal School Training Program is Ineffective or Unavailable: is when resident based formal school training conducted by a formal schoolhouse under TECOM, TRADOC, NETC, AETC, or USCG-FORCECOM is either incorrect, incomplete, insufficient, or unavailable for performance to standard resulting in hazardous conditions or unsafe acts throughout subordinate units or the field / fleet. (Examples include programs that produce officer and enlisted core skills, Military Occupational Specialty, civilian career programs, train-the-trainer programs, operational planning/risk management training, military leadership development, civilian employee supervisor development, staff development, one-time, upgrade or transition programs, combat readiness, etc.)	**OT001**
Distance Learning Training is Ineffective or Unavailable: is when distance learning programs provided by either military service component or contracted organizations is incorrect, incomplete, insufficient, or unavailable for performance to standard resulting in hazardous conditions or unsafe acts throughout subordinate units or the field / fleet. (e.g., simulations, professional military training, career enhancement skills, etc.)	**OT002**
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Standards System Inadequacies

Standards system inadequacies exist when the SIB determined standards, regulations, policies, or procedures, were unclear, impractical, inadequate or did not exist. **Organizational Policy, Procedures, or Processes Issues:** are system inadequacies if flaws in an organization's safety management system (standards, policies, procedural guidance, doctrine, processes, or governance/program management) negatively influenced leader/supervisory or individual performance.	
Question ST1: Did an organizational written standard/policy or lack of such standard/policy at any level create an unsafe situation? (If yes, determine which code(s) influenced the unsafe act, or other system inadequacies, or the severity of damage or injury. If no, finished with Standards System Inadequacy.)	
Organizational Program or Operation not Adequately Assessed: is when the potential risks of a large program, contract management, acquisition program or operations were not assessed adequately, and this inadequacy impacted subordinate level actions.	**OP002**
Provided Unclear, Impractical, or Inadequate Policy, Procedural Guidance or Publications: is when written standards (policies, directives, procedural guidance/standard operating procedures, technical manuals, checklists, or publications) for normal or abnormal/emergency conditions are impractical, too vague/unclear, incorrect or ineffectively disseminated for safe operations throughout the organization or within a subordinate unit, resulting in hazardous conditions or unsafe acts throughout subordinate units or the field / fleet. ***NOTE: Not following a written document that is available, correct, and simple to understand is a supervisory or individual level factor and does not apply to this code.***	**OP003**
Flawed Doctrine/Philosophy: is when the doctrine, philosophy, or concept of operations in an organization is flawed or accepts unnecessary risk which leads to unmitigated hazardous conditions and/or unsafe acts throughout subordinate units or the field / fleet.	**OP005**
Inadequate Program Management / Governance: is when formal programs (e.g., a Program of Record) are implemented without sufficient planning, oversight or support and creates hazardous conditions or unsafe acts throughout subordinate units or the field / fleet. This includes such programs ranging from a specific Occupational Health program to inadequate quality control, original manufacture and rebuild, packaging, assembly of materiel, or the entire safety management system. ***NOTE: This code may be a related to conditions of the "Technological Environment".***	**OP006**
Unit Failed to Provide Clear Written Procedure/Guidance/Policy: is when unit level guidance or policy was discovered to be ineffective, unclear, impractical, or non-existent and resulted in hazardous conditions or unsafe acts. (Examples include standard operating procedures, job/activity hazard analysis, checklists, hazard communication plan, emergency action plan, leave or liberty/pass policies, letter of instruction, operational orders, fragmentary order (FRAGO), concept of operations, etc.)	**SI004**
End of System Inadequacy Tables	

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Notes

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